A Theory of Economic Slack

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CHAPTER 7.

Rigid price norm

In this chapter, we model a realistic price norm and plug into our slackish market model. Building on the evidence presented in chapter 6, the price norm imposes a price that is neither fixed nor flexible but somewhat rigid. We then study how this rigidity shapes the properties of the model—especially the response to shocks.

7.1. Model with rigid prices

In this section we go back to our slackish market model and assume a price norm that encodes how prices move in the real world: they do not fully respond to underlying shocks.

7.1.1. Expression of the rigid price norm

We will now look at what happens when we have a rigid price: a price that moves in the direction of the flexible price, but less than it. To impose this, we specify the following rigid price norm:

$$p^n = \rho \cdot \left(a \cdot k^{-\alpha}\right)^{1-\gamma}, \tag{7.1}$$

where $\gamma \in (0,1]$ determines the amount of rigidity of the price norm and $\rho > 0$ determines the level of the price norm. At $\gamma = 1$, the price is fixed and $p^n = \rho$. At $\gamma = 0$, the price would be flexible. We do not consider $\gamma = 0$ below because we focus on somewhat-rigid prices.

7.1.2. Solution of the model

Next, we solve the model with rigid prices. The tightness that solves the model is given by the supply-equals-demand condition, (5.19), where the price norm is (7.1). Because the price norm is just a function of the parameters, and not of tightness, we can show that the model admits a unique solution, just like we did in the case with a fixed price, and as illustrated in figure 5.3. Indeed, the argument that we presented was valid for any price, so it is valid in particular if the price takes the value given by (7.1).

The solution of the model also continues to be represented graphically by figure 5.3A. What is different is the response to shocks, which we examine next.

7.1.3. Comparative statics

Let us look at a negative demand shock with rigid prices, represented by a reduction in the demand parameter a. We know that for a given price, the market demand is reduced when a falls. But the price also drops with a, which tends to push market demand back up. To determine the combined effect of a, we incorporate the rigid price (7.1) into the market demand (5.15):

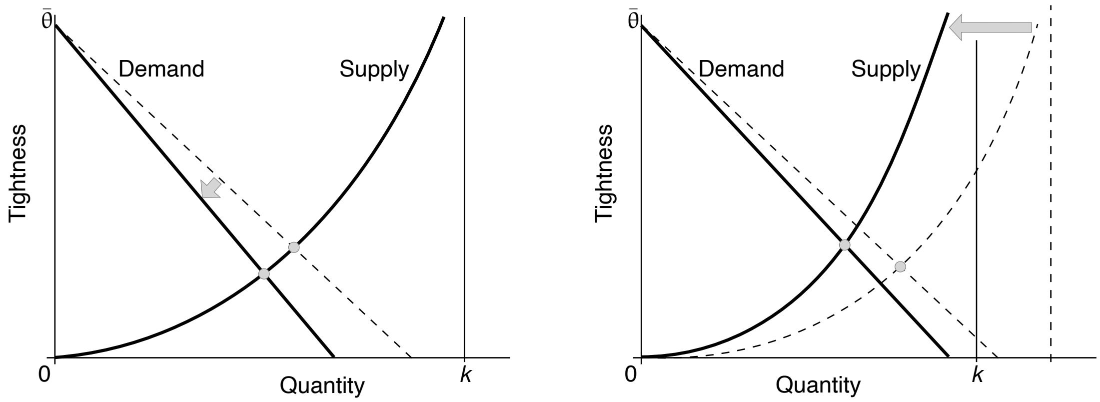
$$y^d(\theta) = \left[\frac{(1-\alpha)a^{\gamma}}{\rho}\right]^{1/\alpha} \cdot \frac{k^{1-\gamma}}{[1+\tau(\theta)]^{1/\alpha-1}}. \tag{7.2}$$

This is a similar equation to the one with fixed prices, except that the parameters appear in a slightly different fashion. In particular, the demand parameter a enters with an exponent γ. Since γ is strictly positive, because of price rigidity, the demand parameter has the exact same effect as with fixed prices, except that the impact of shocks to a will be attenuated by the exponent γ.

So after a negative demand shock, the market demand curve moves inward while the market supply curve remains unmoved, as shown in figure 7.1A. Hence, the comparative statics under demand shocks are the same as with a fixed price. The only difference is quantitative: the size of the effects is slightly attenuated, especially when prices are becoming more flexible (γ closer to 0). Formally, the elasticities of all the variables with respect to a shrink by a factor $\gamma \leq 1$.

Next, let us look at a negative supply shock with rigid prices, represented by a reduction in the market capacity k. The market supply is reduced when k falls. In addition, prices rise, since k appears in the denominator of the price norm (7.1). This increase captures the fact that goods are scarcer so more valuable. That rise in prices will depress the market demand, as illustrated in figure 7.1B.

To perform the comparative statics, we have to determine what happens to tightness, so we have to determine which of the inward movements dominates: the reduction in



A. Negative demand shock

B. Negative supply shock

FIGURE 7.1. Comparative statics in the slackish market model with rigid prices

The market supply is given by (5.5). The market demand is given by (7.2). The price norm is given by (7.1). A negative demand shock is a reduction in the preference for goods a. A negative supply shock is a reduction in market capacity k.

supply (pushing tightness up) or the reduction in demand (pushing tightness down). To do that, we rewrite the supply-equals-demand condition with the expression (7.2) and separate the terms that depend on tightness from the terms that depend on the supply parameter:

$$\left[\frac{(1-\alpha)a^{\gamma}}{\rho}\right]^{1/\alpha} k^{-\gamma} = [1-u(\theta)]\,[1+\tau(\theta)]^{1/\alpha-1}. \tag{7.3}$$

Now, when the capacity k drops, the left-hand side rises, since $-\gamma < 0$. So the right-hand side must rise as well, to maintain the equality. We know that both $1-u(\theta)$ and $1+\tau(\theta)$ are increasing functions of θ, and $1/\alpha > 1$, so the right-hand side is an increasing function of θ. This tells us that θ must rise when k falls. So just as with fixed prices, tightness goes up after a negative supply shock. As with demand shocks, the response of tightness is attenuated when prices are less rigid (lower γ). All the other variables respond as in the case with fixed prices, except of course the market price which rises here instead of remaining fixed.

7.2. Comparing fixed, rigid, and flexible prices

Overall, we have seen in the previous section that all the comparative statics remain the same if the market price is not completely fixed but simply rigid (table 7.1). It is only when prices are flexible that comparative statics completely change.

This can be seen in (7.3): for any $\gamma \in (0,1]$, including the case of fixed prices ($\gamma = 1$), tightness responds the same to demand and supply shocks. But for flexible prices ($\gamma = 0$), the terms a^{γ} and $k^{-\gamma}$ drop out, so tightness does not respond to either demand or supply shocks. In that case the shocks are entirely absorbed by prices, so tightness does not change.

An implication is that demand shocks are entirely neutral under flexible prices. Because tightness remains the same, the other quantities (output, consumption, rate of slack, visits) also remain the same.

Additionally, supply shocks do not affect tightness under flexible prices, although they do affect some quantities. Because tightness does not change, the rate of slack $u(\theta)$ does not change. However, because capacity has decreased, output $y = [1 - u(\theta)]k$ decreases, consumption $c = y/[1 + \tau(\theta)]$ decreases, and visits $v = y/q(\theta)$ decrease too.

Why is the response so different under flexible prices, compared to rigid prices? Consider a negative demand shock, represented by a fall in demand parameter a. Under a flexible price, the price norm moves proportionally with a, leaving the market demand curve unchanged and therefore leaving tightness unchanged. This is because the flexible price decreases enough to exactly offset the fall in marginal utility and keep the market demand curve in the same position. We can see this in the expression for the flexible price, by setting $\gamma = 0$ in the price norm (7.1). The demand parameter a is in the numerator, so the price drops in proportion to a after the shock. Hence the price drops as much as the marginal utility of consumption, which neutralizes that drop and brings back the market demand to its original position. After incorporating the drop in price, the market demand is unchanged, so tightness is unchanged.

A similar logic explains why flexible prices also neutralize shocks to market capacity. Consider a negative supply shock, represented by a fall in market capacity k. For a given tightness, lower capacity reduces output and consumption. Under flexible prices, this reduction in consumption raises the marginal utility of consumption. Because the flexible price is proportional to marginal utility, the price increases exactly enough to reflect the increased scarcity of goods. This increase in the price shifts the market demand curve inward just enough to offset the inward shift of the market supply curve. The two effects cancel out, leaving market tightness unchanged. Formally, this can be seen by setting $\gamma = 0$ in the price norm (7.1), which implies that the price rises proportionally when k falls. Under flexible prices, supply shocks are therefore fully absorbed by prices, so tightness does not respond.

7.3. A possible source of price flexibility: bargaining

The price norm (7.1) introduces a flexible benchmark, which moves in step with demand and supply parameters so as to absorb supply and demand shocks. A natural question is: How could this flexible benchmark arise in the market? In this section, we show that flexible prices appear when sellers and buyers bargain over prices.

TABLE 7.1. Comparative statics in the slackish market model for various price norms

	Sales y	Tightness θ	Price p	Slack rate u	Consumption c	Visits v
A. Fixed price norm: $\gamma = 1$						
Decrease in demand a	–	–	0	+	?	–
Decrease in supply k	–	+	0	–	–	?
B. Rigid price norm: $\gamma \in (0, 1)$						
Decrease in demand a	–	–	–	+	?	–
Decrease in supply k	–	+	+	–	–	?
C. Flexible price norm: $\gamma = 0$						
Decrease in demand a	0	0	–	0	0	0
Decrease in supply k	–	0	+	0	–	–

The comparative statics are obtained from equation (7.3) and are illustrated in figures 5.4 and 7.1. A variable's increase is denoted by "+", a decrease by "-", no response by "0", and an indeterminate response by "?".

7.3.1. Outcome of the bargaining process

Following Diamond (1982), we assume that the outcome of the bargaining process is that buyer and seller share the surplus of their trade. This means that the seller surplus $\mathcal{S}$ amounts to a share χ of the total surplus $\mathcal{T}$ and the buyer surplus $\mathcal{B}$ amounts to a share $1 - \chi$ of the total surplus, where $\chi \in (0, 1)$ is the bargaining power of the seller.

In section 6.2, we established that for a trade at price p, the seller's surplus is just the price, $\mathcal{S} = p$, and the total surplus is just the marginal utility from consumption, $\mathcal{T} = (1 - \alpha)ac^{-\alpha}$. Since the surplus-sharing solution requires that $\mathcal{S} = \chi \cdot \mathcal{T}$, the bargained price is

$$p = \chi \cdot (1 - \alpha)ac^{-\alpha}. \tag{7.4}$$

That is, through bargaining, sellers keep a share χ of the trade surplus, which is just the marginal utility from the good enjoyed by buyers. Buyers are left with a fraction $1 - \chi$ of the surplus. They enjoy the full marginal utility by consuming the good but must pay a share χ of the marginal utility to sellers; so they are left with $1 - \chi$ of the marginal utility in the end. If sellers have all the bargaining power ($\chi = 1$), the price is just the marginal utility. If buyers have all the bargaining power ($\chi = 0$), the price is 0.

In chapter 5, we noted that all price norms reduce to functions of market tightness. This is true here too. In the model, consumption and output are related by $c = y/[1 + \tau(\theta)]$, and output and tightness are related by $y = [1 - u(\theta)]k$, so consumption can be written a function of tightness too:

$$c = \frac{1 - u(\theta)}{1 + \tau(\theta)} \cdot k. \tag{7.5}$$

By plugging this expression into (7.4), we write the bargained price as a function of tightness:

$$p = \chi(1-\alpha)ak^{-\alpha}\left[\frac{1-u(\theta)}{1+\tau(\theta)}\right]^{-\alpha}. \tag{7.6}$$

7.3.2. Solution with bargained prices

Solving the model with bargaining is a bit different because the market demand takes a very special form once it is combined with the bargained price. When we derived the market demand in chapter 5, we obtained it by equalizing the marginal utility of consumption to the marginal cost of consumption: $(1-\alpha)ac^{-\alpha} = [1+\tau(\theta)]p$. And we have just seen that the bargained price satisfies $p = \chi(1-\alpha)ac^{-\alpha}$. Combining both, we see that once we incorporate the bargained price into the market demand, the market demand becomes degenerate. It pins down a unique tightness irrespective of quantities traded: $1 = [1+\tau(\theta)]\chi$. This condition on tightness can be rewritten as a condition on the matching wedge:

$$\tau(\theta) = \frac{1-\chi}{\chi}. \tag{7.7}$$

In the tightness-quantity diagram, the market demand is therefore horizontal:

$$\theta = \tau^{-1}\left(\frac{1-\chi}{\chi}\right), \tag{7.8}$$

where τ^{-1} is the inverse of the matching wedge. The inverse is well defined on $(0,\infty)$ because τ is strictly increasing on $(0,\bar{\theta}) \to (0,\infty)$. The inverse itself is strictly increasing.

7.3.3. Flexibility of bargained prices

From the expression (7.8) for market tightness, we can express the bargained price (7.6) as a function of the parameters of the model. The bargained price is given by

$$p = (1-\alpha)\chi^{1-\alpha}\left[1-u\left(\tau^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{-\alpha} ak^{-\alpha}. \tag{7.9}$$

We simplified the expression using the fact that when (7.8) holds, $1+\tau(\theta) = 1/\chi$.

We now see that if we set our price norm (7.1) with $\gamma = 0$ and

$$\rho = (1-\alpha)\chi^{1-\alpha}\left[1-u\left(\tau^{-1}\left(\frac{1-\chi}{\chi}\right)\right)\right]^{-\alpha},$$

we obtain the bargained price described by (7.9). Hence, bargaining provides a foundation for a flexible price.

Hence, price bargaining provide an explanation for price movements in the direction of flexibility. Bargained prices are too flexible to be realistic: they prevent the rate of slack from changing, which is not consistent with the real world, where the amount of slack in markets varies considerably over the business cycle (chapter 2). But with a small amount of bargaining in markets, prices might be not completely fixed but somewhat flexible towards the bargained price. Thus, price bargaining might provide a foundation for the price norm (7.1).

7.4. Quantifying the response of the model to shocks

Comparative statics tell us in which directions the model variables respond to demand and supply shocks. But it is also possible to determine the amplitude of the response of the variables. The key step is to compute the elasticity of market tightness with respect to the parameters at the source of the shocks (a and k). Given that tightness determines all other variables in the model, we can quantify the response of any variable from the response of tightness. From the elasticity of tightness, we will see in particular that the response of the model to shocks is determined critically by the rigidity of prices. Throughout the section, we use the elasticity results introduced in appendix B.

7.4.1. Response to demand shocks

We start with the elasticity of tightness with respect to the demand parameter a. This elasticity tells us how much the model responds to demand shocks. We start from the supply-equals-demand equation, which implicitly gives tightness: $y^d(\theta, a) = y^s(\theta)$, where the market demand is given by (7.2) and the market supply takes the usual form, (5.5). Using implicit differentiation with elasticities, we get

$$\epsilon_a^d + \epsilon_\theta^d \cdot \epsilon_a^\theta = \epsilon_\theta^s \cdot \epsilon_a^\theta,$$

where ϵ_a^d is the partial elasticity of market demand y^d with respect to the demand parameter a, ϵ_θ^d is the partial elasticity of market demand y^d with respect to tightness θ, ϵ_θ^s is the elasticity of market supply y^s with respect to tightness θ, and ϵ_a^θ is the elasticity of tightness θ with respect to the demand parameter a—which is the elasticity that we are aiming to compute.

Refactoring this condition, we isolate the elasticity of tightness with respect to market demand:

$$\epsilon_a^\theta = \frac{\epsilon_a^d}{\epsilon_\theta^s - \epsilon_\theta^d}. \tag{7.10}$$

This expression shows that the response of tightness to a demand shock depends of course on the response of demand to the shock (ϵ_a^d), and critically on the relative slopes of the demand and supply curves ($\epsilon_\theta^s - \epsilon_\theta^d$).

Next, we compute the individual elasticities involved in (7.10). Using the elasticities (4.7) and (4.8), we get

$$\epsilon_a^d = \frac{\gamma}{\alpha}, \quad \epsilon_\theta^d = -\frac{1-\alpha}{\alpha} \cdot \epsilon_\theta^{1+\tau}, \quad \epsilon_\theta^s = 1 - \eta(\theta), \tag{7.11}$$

where $\epsilon_\theta^{1+\tau}$ is the elasticity of $1+\tau$ with respect to tightness θ. Given (4.7) and (5.8), we infer that the elasticity of $1+\tau(\theta) = q(\theta)/[q(\theta) - \kappa]$ with respect to θ is

$$\epsilon_\theta^{1+\tau} = [-\eta(\theta)] - [-\eta(\theta)] \cdot \frac{q(\theta)}{q(\theta) - \kappa} = \eta(\theta)\tau(\theta). \tag{7.12}$$

Hence, the elasticity of the market demand with respect to market tightness is

$$\epsilon_\theta^d = -\frac{1-\alpha}{\alpha} \cdot \eta(\theta)\tau(\theta). \tag{7.13}$$

Plugging in (7.10) the values of the elasticities, we obtain:

$$\epsilon_a^\theta = \frac{\gamma}{[1-\eta(\theta)]\,\alpha + (1-\alpha)\eta(\theta)\tau(\theta)}. \tag{7.14}$$

This expression highlights several properties of the response of the market to demand shocks. First, tightness goes up when demand is stronger ($\epsilon_a^\theta > 0$). Second, if prices are flexible ($\gamma = 0$), then tightness does not respond to demand shocks ($\epsilon_a^\theta = 0$), which means that the market does not respond to demand shocks—in line with what we saw when we looked at bargained prices (table 7.1). Third, the more rigid prices are (higher γ), the stronger is the response of tightness, and thus of the other variables, to demand shocks.

7.4.2. Response to supply shocks

Next we move to the elasticity of tightness with respect to the market capacity k, which determines the market supply. This elasticity tells us how much the model responds to supply shocks. We start again from the supply-equals-demand equation, which implicitly gives tightness: $y^d(\theta, k) = y^s(\theta, k)$, where the market demand is given by (7.2) and the market supply takes the usual form, (5.5). Using implicit differentiation with elasticities, we get

$$\epsilon_\theta^d \cdot \epsilon_k^\theta + \epsilon_k^d = \epsilon_\theta^s \cdot \epsilon_k^\theta + \epsilon_k^s,$$

where ϵ_k^s is the partial elasticity of market supply y^s with respect to the market capacity k and ϵ_k^θ is the elasticity of tightness θ with respect to the market capacity k—which is the elasticity that we are aiming to compute here. Besides the elasticity above, we also have

$$\epsilon_k^d = 1 - \gamma, \quad \epsilon_k^s = 1.$$

Combining these results, we obtain first that

$$\epsilon_k^\theta = \frac{\epsilon_k^d - \epsilon_k^s}{\epsilon_\theta^s - \epsilon_\theta^d}, \tag{7.15}$$

which shows that the response of tightness to a supply shock depends on the response of demand and supply to the shock (ϵ_k^s and ϵ_k^d), and critically on the slopes of the demand and supply curves ($\epsilon_\theta^s - \epsilon_\theta^d$). Next, we can plug in this abstract expression the various values of the elasticities and we obtain:

$$\epsilon_k^\theta = \frac{-\gamma}{1 - \eta + \frac{1-\alpha}{\alpha} \cdot \eta\tau(\theta)}. \tag{7.16}$$

From (7.16) we see several properties of the response of the market to supply shocks. First, tightness goes down when supply is higher ($\epsilon_k^\theta < 0$). Second, if prices are flexible ($\gamma = 0$), tightness does not respond to supply shocks ($\epsilon_k^\theta = 0$). Third, the response of tightness to supply shocks is stronger when prices are more rigid (higher γ).

7.4.3. State dependence

A critical property of the slackish model emerges from the elasticity formulas: the model exhibits state dependence. This means that the model behaves differently when the market is slack or tight. To display such state dependence, we now compute the response of output to demand and supply shocks.

We begin by picking a relationship that relates output to tightness. A convenient choice is the market supply: $y = y^s(\theta, k)$. From this relationship, we then obtain $\epsilon_a^y = \epsilon_\theta^s \cdot \epsilon_a^\theta$ so

$$\epsilon_a^y = \frac{\gamma}{\alpha + \frac{\eta(\theta)}{1-\eta(\theta)} \cdot (1-\alpha)\tau(\theta)}.$$

So the response of output to demand shocks has broadly the same properties as the response of tightness.

In addition, we see here that the response of output to demand shocks is stronger in a slacker market. Indeed, the matching wedge $\tau(\theta)$ is highly procyclical. Moreover, by the assumption from chapter 4 that the matching elasticity $\eta(\theta)$ is increasing in tightness, the ratio $\eta(\theta)/[1-\eta(\theta)]$ is also increasing in tightness. Combined with the strict increase of $\tau(\theta)$, this implies that the term $\eta(\theta)\tau(\theta)/[1-\eta(\theta)]$ is strictly increasing in tightness. Thus, since all other terms in the elasticity are constant, the elasticity ϵ_a^y is decreasing in tightness. This means that when the market is slack, the elasticity is high: output responds sharply to changes in market demand. By contrast, when the market is tight, the elasticity is low: output does not respond much to changes in market demand.

For supply shocks, we have $\epsilon_k^y = \epsilon_k^s + \epsilon_\theta^s \cdot \epsilon_k^\theta$ so

$$\epsilon_k^y = 1 - \frac{\gamma}{1 + \frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{1-\alpha}{\alpha} \cdot \tau(\theta)}.$$

Since $\gamma \leq 1$, the elasticity is strictly positive ($\epsilon_k^y > 0$). So unlike tightness, which falls after an increase in supply, output rises after an increase in supply. Furthermore, output rises more when prices are less rigid. With a flexible price ($\gamma = 0$), output moves one-for-one with capacity ($\epsilon_k^y = 1$); when prices are rigid, the move is less than one-for-one ($\epsilon_k^y < 1$).

We also see that the response of output to supply shocks is stronger in a tighter market. Following the same logic as above, we know that the term $\eta(\theta)\tau(\theta)/[1 - \eta(\theta)]$ is strictly increasing in tightness. This result implies that the elasticity ϵ_k^y is increasing in tightness. This means that when the market is slack, the elasticity is low: output does not respond much to changes in market capacity. By contrast, when the market is tight, the elasticity is high: output responds significantly to changes in market capacity.

We have just established that the slackish model is state-dependent: the response of output to shocks depends on the state of the market. The reason behind state dependence is the nonlinearity of the slackish model. Because of the curvature of the market demand and supply curves, shocks have systematically different effects when tightness is high or low. We will come back to this result in the dynamic model, where state dependence is exacerbated by a stronger nonlinearity (chapter 9).

7.5. Summary

In this chapter we incorporate price rigidity into our slackish market model. Prices respond partially to shocks with some rigidity parameter $\gamma \in (0, 1]$, so adjustment to shocks is between the extremes of fixed and flexible prices. Both prices and tightness adjust, with the elasticity of tightness to demand and supply shocks determined by γ. This price rigidity explains why slack varies over the business cycle—a central feature of real markets that models with flexible prices cannot capture.

Because of the curvature of the market supply, the slackish model is state dependent. Output responds more to demand shocks in slack conditions than in tight conditions, and more to supply shocks in tight conditions than in slack conditions.

Bibliography

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